Administrarea privilegiilor
(Cap. 16)

1. Acordare privilegii pe o tabela din userul curent unui alt user

```
SQL> connect scott@bd /tiger

SQL> grant select, insert,  update  on emp to test;
```

Grant succeeded.

2. Verificare privilegii create

```
SQL> connect test@bd /test
SQL> select * from scott.emp where deptno=10 ;
SQL> insert into scott.emp values (1111,'Test','Student',7902,sysdate, 1000, 200, 10);
SQL> update scott.emp set sal=1500 where empno=1111;
```

3. Crearea unei copii pentru o tabela

```
SQL> connect scott@bd /tiger
SQL> grant create table to test;
SQL> create table emp_copy as select * from emp;
SQL> grant select, insert,  update  on emp_copy to test;

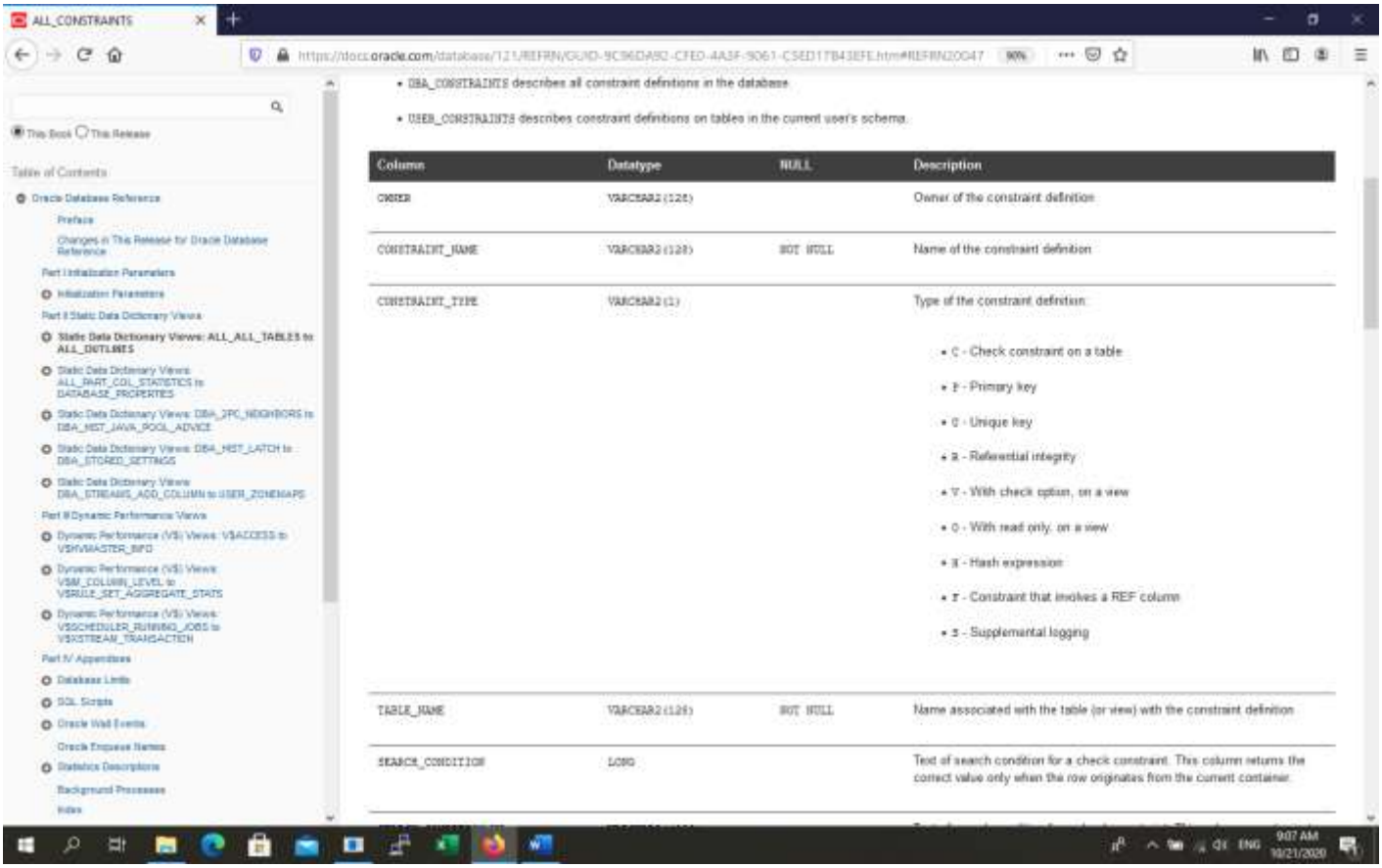

SQL> connect test@bd /test
SQL> create table emp_copy as select * from scott.emp;
```

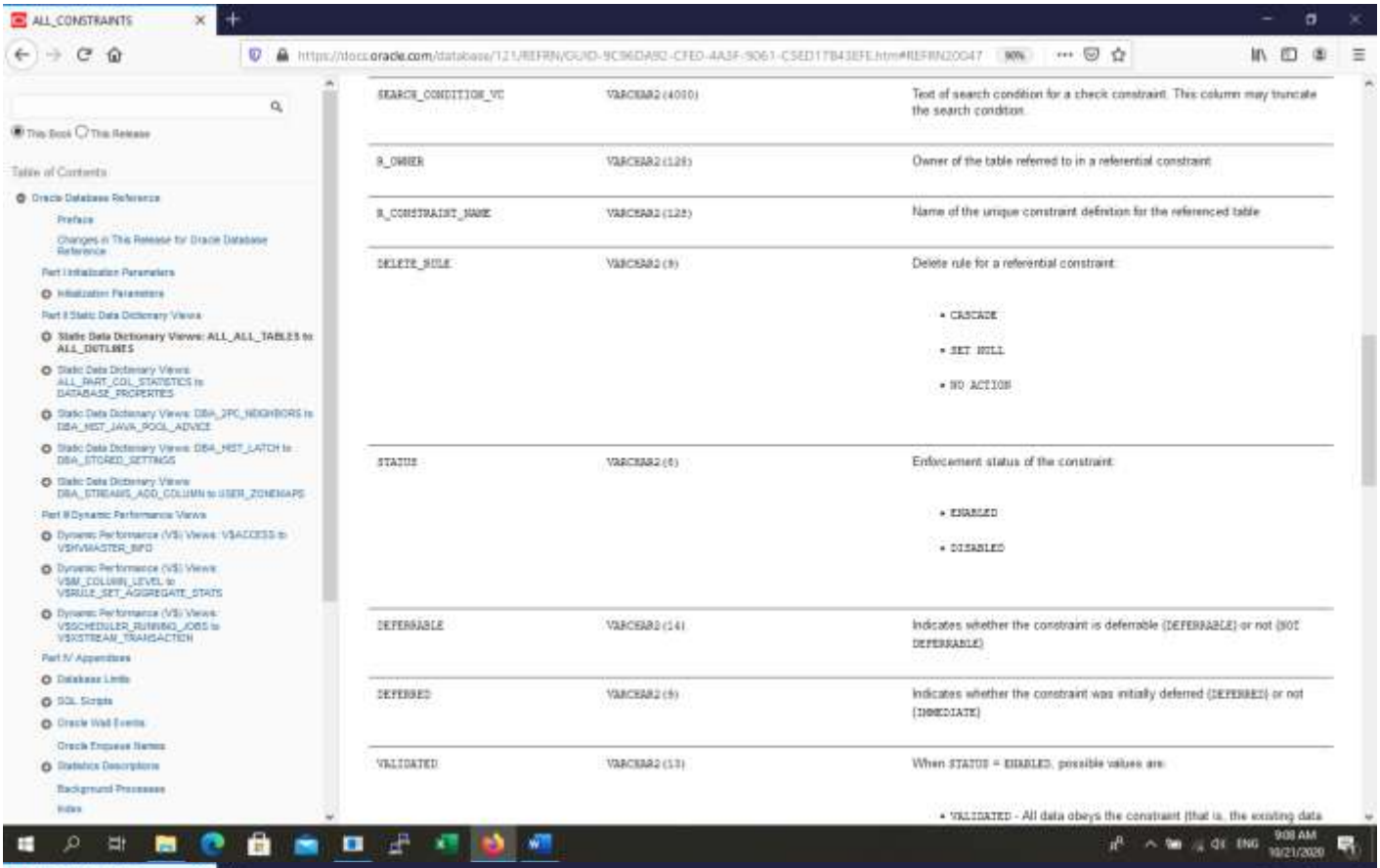
Obs. Cand se creeaza o tabela nu se creeaza automat si constrangerile de integritate :

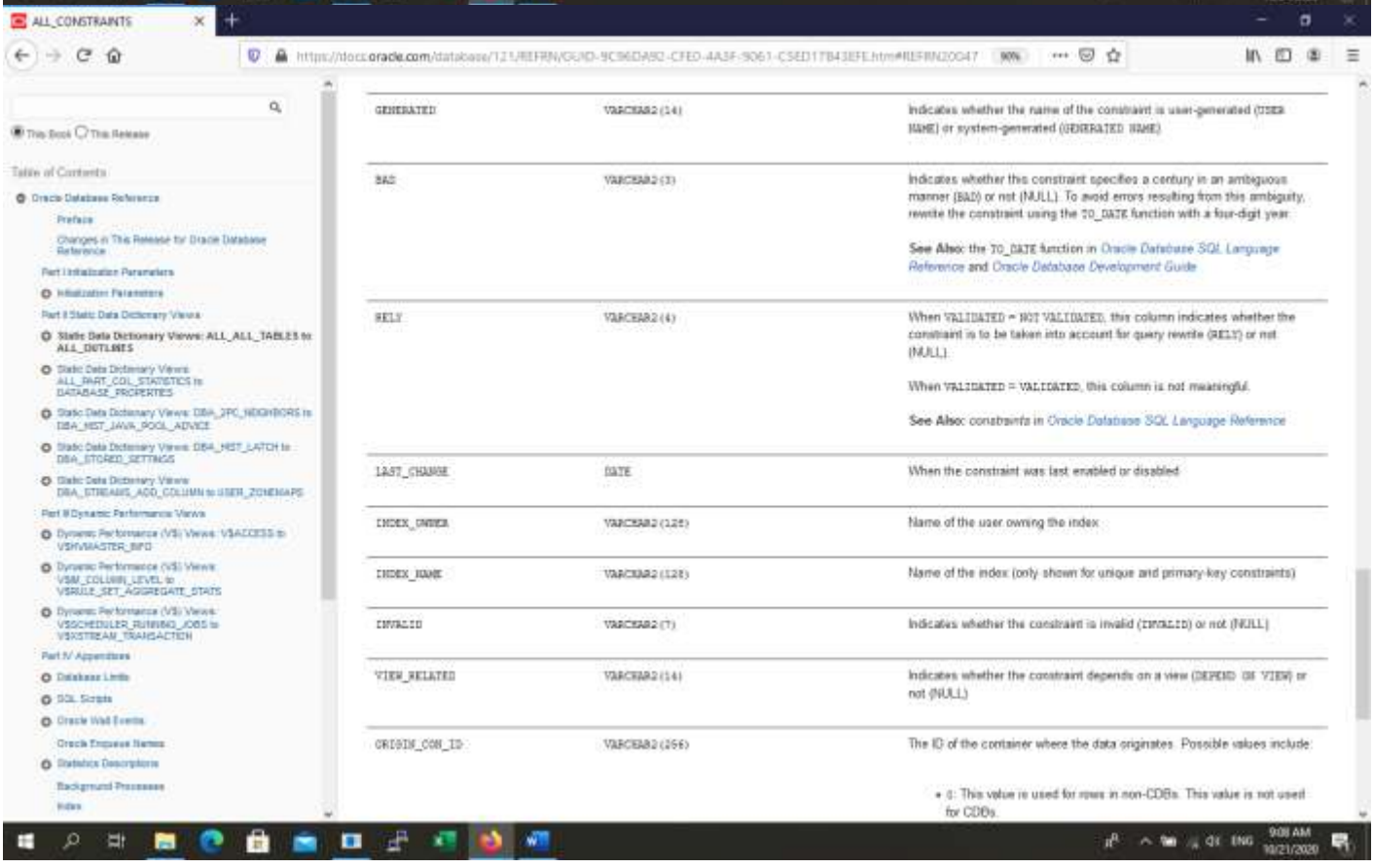
```
SQL> desc USER_CONSTRAINTS
```

Name	Null?	Type
OWNER		VARCHAR2(128)
CONSTRAINT_NAME		VARCHAR2(128)
CONSTRAINT_TYPE		VARCHAR2(1)
TABLE_NAME		VARCHAR2(128)
SEARCH_CONDITION		LONG
SEARCH_CONDITION_VC		VARCHAR2(4000)
R_OWNER		VARCHAR2(128)
R_CONSTRAINT_NAME		VARCHAR2(128)
DELETE_RULE		VARCHAR2(9)
STATUS		VARCHAR2(8)
DEFERRABLE		VARCHAR2(14)
DEFERRED		VARCHAR2(9)
VALIDATED		VARCHAR2(13)

GENERATED	VARCHAR2(14)
BAD	VARCHAR2(3)
RELY	VARCHAR2(4)
LAST_CHANGE	DATE
INDEX_OWNER	VARCHAR2(128)
INDEX_NAME	VARCHAR2(128)
INVALID	VARCHAR2(7)
VIEW_RELATED	VARCHAR2(14)
ORIGIN_CON_ID	NUMBER




ALL_CONSTRAINTS
SEARCH_CONDITION_VC
VARCHAR2(4000)
Text of search condition for a check constraint. This column may truncate the search condition
R_OWNER
VARCHAR2(128)
Owner of the table referred to in a referential constraint
R_CONSTRAINT_NAME
VARCHAR2(128)
Name of the unique constraint definition for the referenced table
DELETE_RULE
VARCHAR2(9)
Delete rule for a referential constraint:
CASCADE
SET NULL
NO ACTION
STATUS
VARCHAR2(8)
Enforcement status of the constraint:
ENABLED
DISABLED
DEFERRABLE
VARCHAR2(14)
Indicates whether the constraint is deferrable (DEFERRABLE) or not (NOT DEFERRABLE)
DEFERRED
VARCHAR2(9)
Indicates whether the constraint was initially deferred (DEFERRED) or not (IMMEDIATE)
VALIDATED
VARCHAR2(13)
When STATUS = ENABLED, possible values are:
VALIDATED - All data obeys the constraint (that is, the existing data


ALL_CONSTRAINTS
GENERATED
VARCHAR2(14)
Indicates whether the name of the constraint is user-generated (USER NAME) or system-generated (GENERATED NAME)
BAD
VARCHAR2(3)
Indicates whether this constraint specifies a century in an ambiguous manner (BAD) or not (NULL). To avoid errors resulting from this ambiguity, rewrite the constraint using the TO_DATE function with a four-digit year.
See Also: the TO_DATE function in Oracle Database SQL Language Reference and Oracle Database Development Guide
RELY
VARCHAR2(4)
When VALIDATED = NOT VALIDATED, this column indicates whether the constraint is to be taken into account for query rewrite (RELY) or not (NULL)
When VALIDATED = VALIDATED, this column is not meaningful.
See Also: constraints in Oracle Database SQL Language Reference
LAST_CHANGE
DATE
When the constraint was last enabled or disabled
INDEX_OWNER
VARCHAR2(128)
Name of the user owning the index
INDEX_NAME
VARCHAR2(128)
Name of the index (only shown for unique and primary-key constraints)
INVALID
VARCHAR2(7)
Indicates whether the constraint is invalid (INVALID) or not (NULL)
VIEW_RELATED
VARCHAR2(14)
Indicates whether the constraint depends on a view (DEPEND ON VIEW) or not (NULL)
ORIGIN_CON_ID
VARCHAR2(256)
The ID of the container where the data originates. Possible values include:
0: This value is used for rows in non-CDBs. This value is not used for CDBs.

SQL> connect scott@bd /tiger
SQL> select owner,table_name,constraint_name from user_constraints where owner='SCOTT'
and table_name=’EMP’;

OWNER	TABLE_NAME	CONSTRAINT_NAME
SCOTT	EMP	PK_EMP
SCOTT	EMP	FK_DEPTNO

SQL> connect test@bd /test
SQL> select owner,table_name,constraint_name from user_constraints where owner='TEST'
and table_name=’EMP_COPY’;

No rows selected.

4. Revocarea unui privilegiu

SQL> connect scott@bd /tiger

SQL> revoke select on emp_copy from test;

Revoke succeeded.

SQL> connect test@bd /test

SQL> select * from scott.emp_copy;

ERROR at line 1:
ORA-01031: insufficient privileges

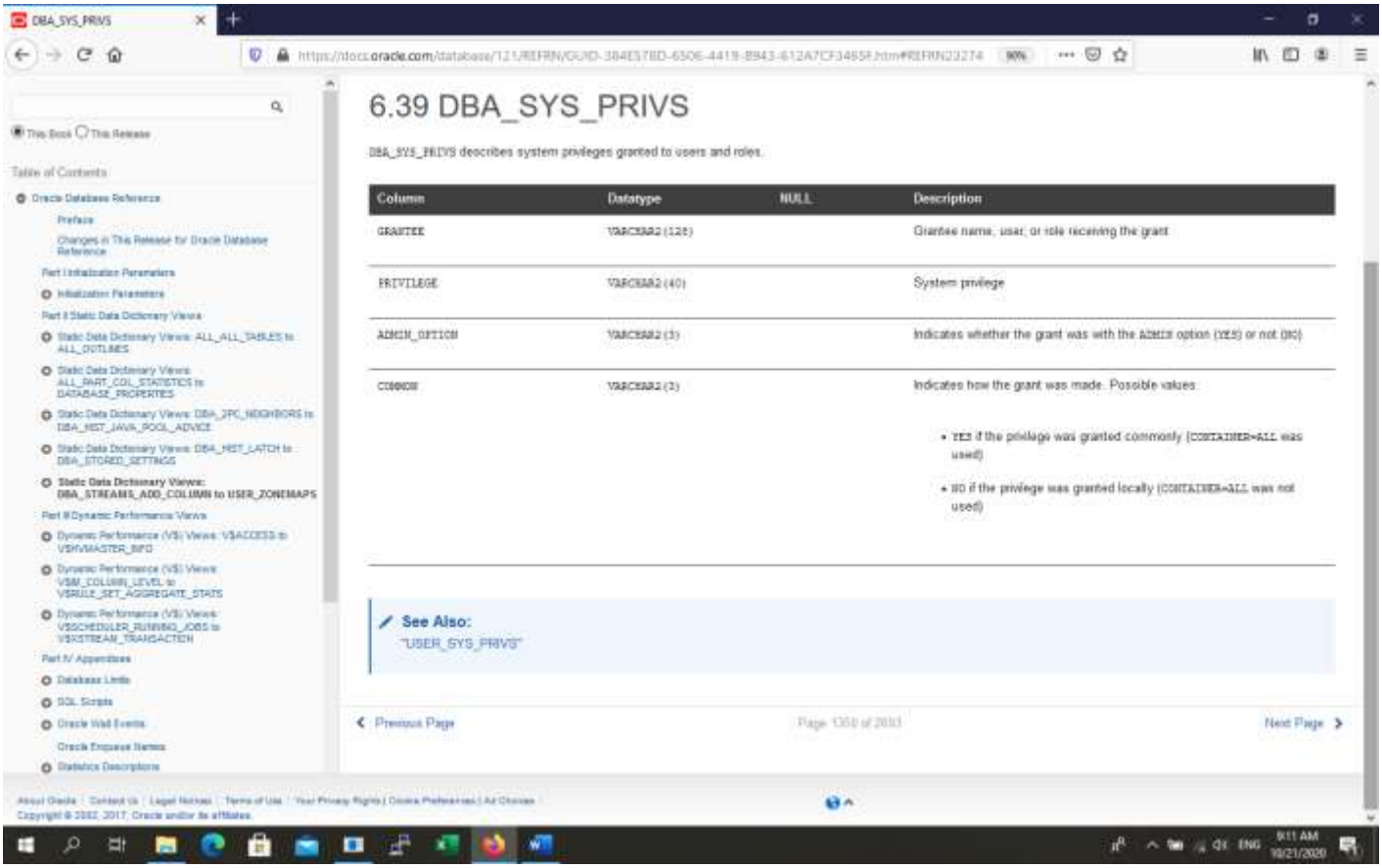
SQL> connect scott@bd /tiger

SQL> revoke all on emp_copy from test cascade constraints; -- revocarea tuturor privilegiilor ; se sterg si toate constrangerile de integritate create de user folosind REFERENCES sau ALL.

5. Vizualizarea din dictionar a privilegiilor de sistem acordate unui user

SQL> connect scott@bd /tiger
SQL> desc DBA_SYS_PRIVS

Name	Null?	Type
GRANTEE	NOT NULL	VARCHAR2(30)
PRIVILEGE	NOT NULL	VARCHAR2(40)
ADMIN_OPTION		VARCHAR2(3)
COMMON		VARCHAR2(3)



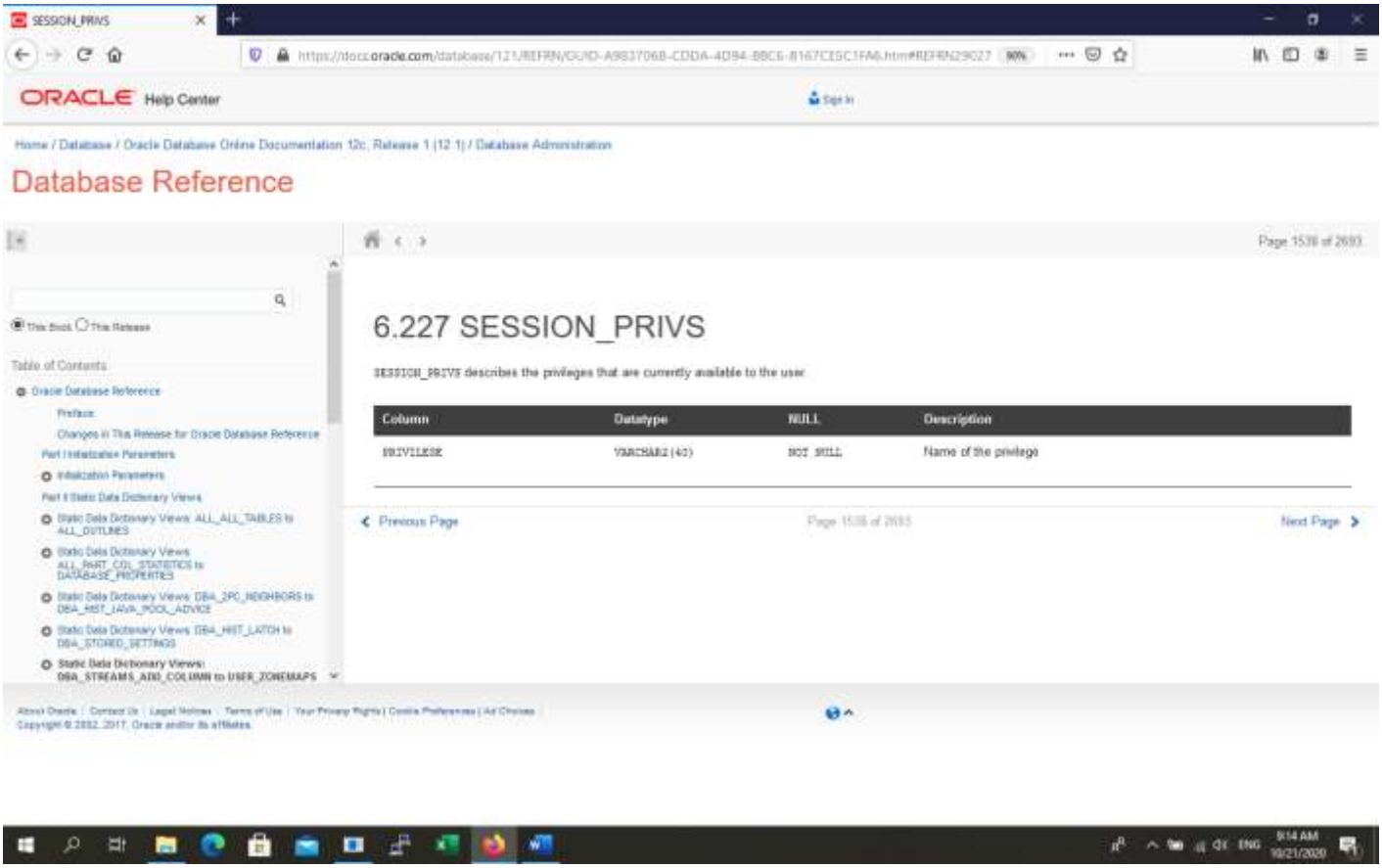
SQL> select * from DBA_SYS_PRIVS where grantee='TEST'

GRANTEE	PRIVILEGE	ADMIN
TEST	CREATE TABLE	NO
TEST	CREATE SESSION	NO

6. Vizualizarea din dictionar a tuturor privilegiilor de system acordate userului curent

SQL> connect scott@bd /tiger
SQL> desc SESSION_PRIVS

Name	Null?	Type
PRIVILEGE	NOT NULL	VARCHAR2(40)



SQL> select * from SESSION_PRIVS order by privilege;

PRIVILEGE

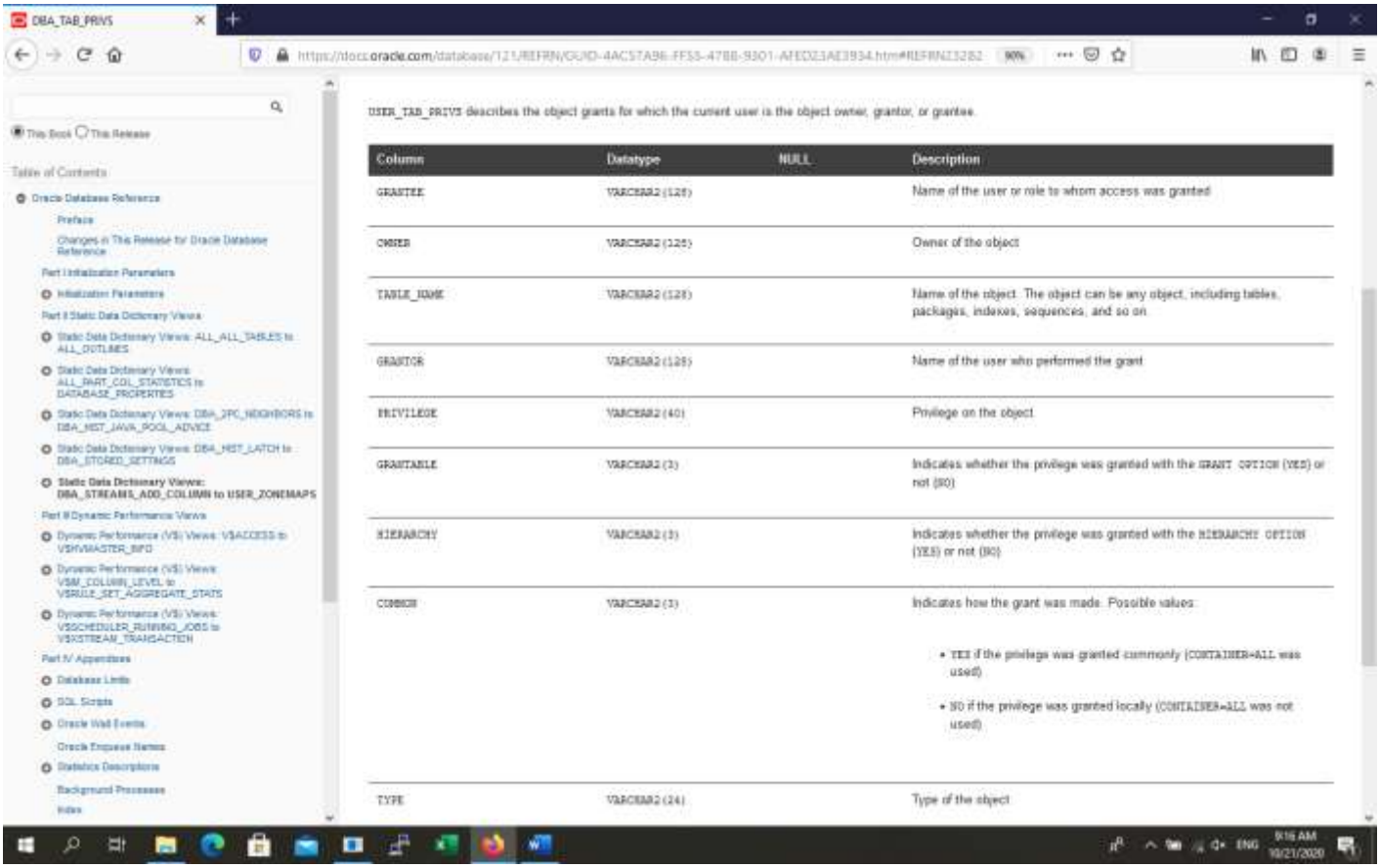
ALTER SYSTEM
AUDIT SYSTEM
CREATE SESSION
ALTER SESSION
RESTRICTED SESSION
CREATE TABLESPACE
ALTER TABLESPACE
MANAGE TABLESPACE
DROP TABLESPACE
UNLIMITED TABLESPACE
CREATE USER
. . . .

7. Vizualizarea din dictionar a tuturor privilegiilor acordate pe obiectele bazei de date

SQL> connect scott@bd /tiger
SQL> desc DBA_TAB_PRIVS

Name Null? Type

GRANTEE	NOT NULL	VARCHAR2(30)
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
GRANTOR	NOT NULL	VARCHAR2(30)
PRIVILEGE	NOT NULL	VARCHAR2(40)
GRANTABLE		VARCHAR2(3)
HIERARCHY		VARCHAR2(3)
COMMON		VARCHAR2(3)
TYPE		VARCHAR2(24)



SQL> select grantee, owner, table_name, grantor, privilege from DBA_TAB_PRIVS where grantee='TEST';

GRANTEE	OWNER	TABLE_NAME	GRANTOR	PRIVILEGE
TEST	SCOTT	EMP	SCOTT	INSERT
TEST	SCOTT	EMP	SCOTT	UPDATE

8. Privilegii acordate pe anumite coloane ale unei tabele (numai insert si update)

SQL> grant insert(empno, ename) on emp to test;
SQL> grant update(empno, ename) on emp to test;

SQL> conn test

```
SQL> insert into scott.emp(empno,ename) values (1111,'Popa');

1 row created.

SQL> insert into scott.emp(empno,sal) values (1111,100);
insert into scott.emp(empno,sal) values (1111,100)
                  *
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> update scott.emp set ename='Tache' where ename='Popa';

1 row updated.

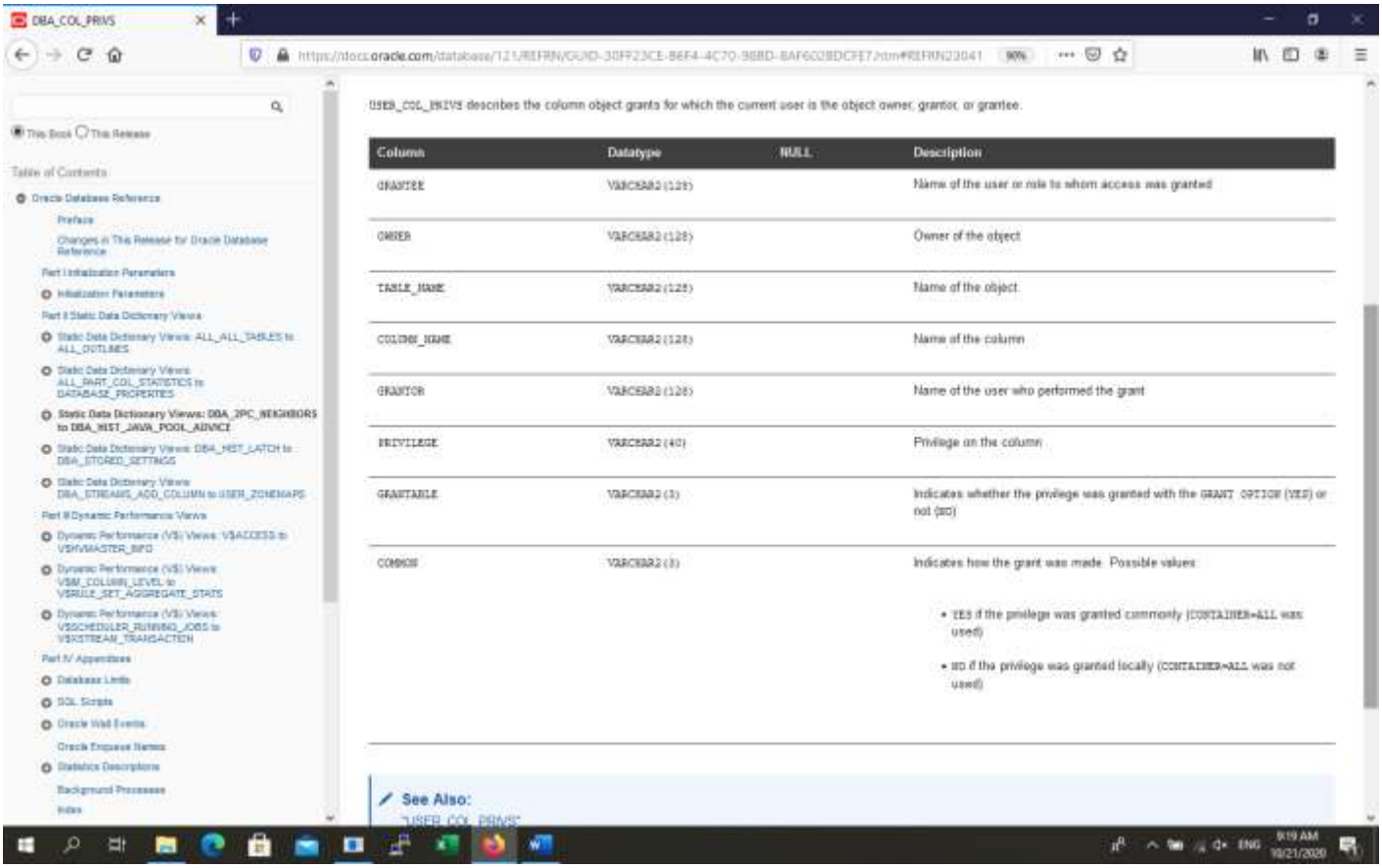
SQL> update scott.emp set sal=1000 where ename='Tache';

ERROR at line 1:
ORA-01031: insufficient privileges
```

9. Vizualizare privilegii acordate pe anumite coloane

```
SQL> desc DBA_COL_PRIVS

Name                                      Null?    Type
----------------------------------------- -------- ----------------
GRANTEE                                   NOT NULL VARCHAR2(30)
OWNER                                     NOT NULL VARCHAR2(30)
TABLE_NAME                                NOT NULL VARCHAR2(30)
COLUMN_NAME                               NOT NULL VARCHAR2(30)
GRANTOR                                   NOT NULL VARCHAR2(30)
PRIVILEGE                                 NOT NULL VARCHAR2(40)
GRANTABLE                                          VARCHAR2(3)
COMMON                                             VARCHAR2(3)
```



SQL> select * from DBA_COL_PRIVS where grantor='SCOTT';

GRANTEE	OWNER	TABLE_NAME	COLUMN_NAME	GRANTOR	PRIVILEGE
TEST	SCOTT	EMP	ENAME	SCOTT	UPDATE
TEST	SCOTT	EMP	EMPNO	SCOTT	UPDATE

10. Creare privilegii de conectare pentru un user si vizualizare privilegii

SQL> conn sys as sysdba
.
SQL> grant connect to test;

Grant succeeded.

SQL> grant create session to test;

Grant succeeded.

SQL> alter system enable restricted session;

System altered.

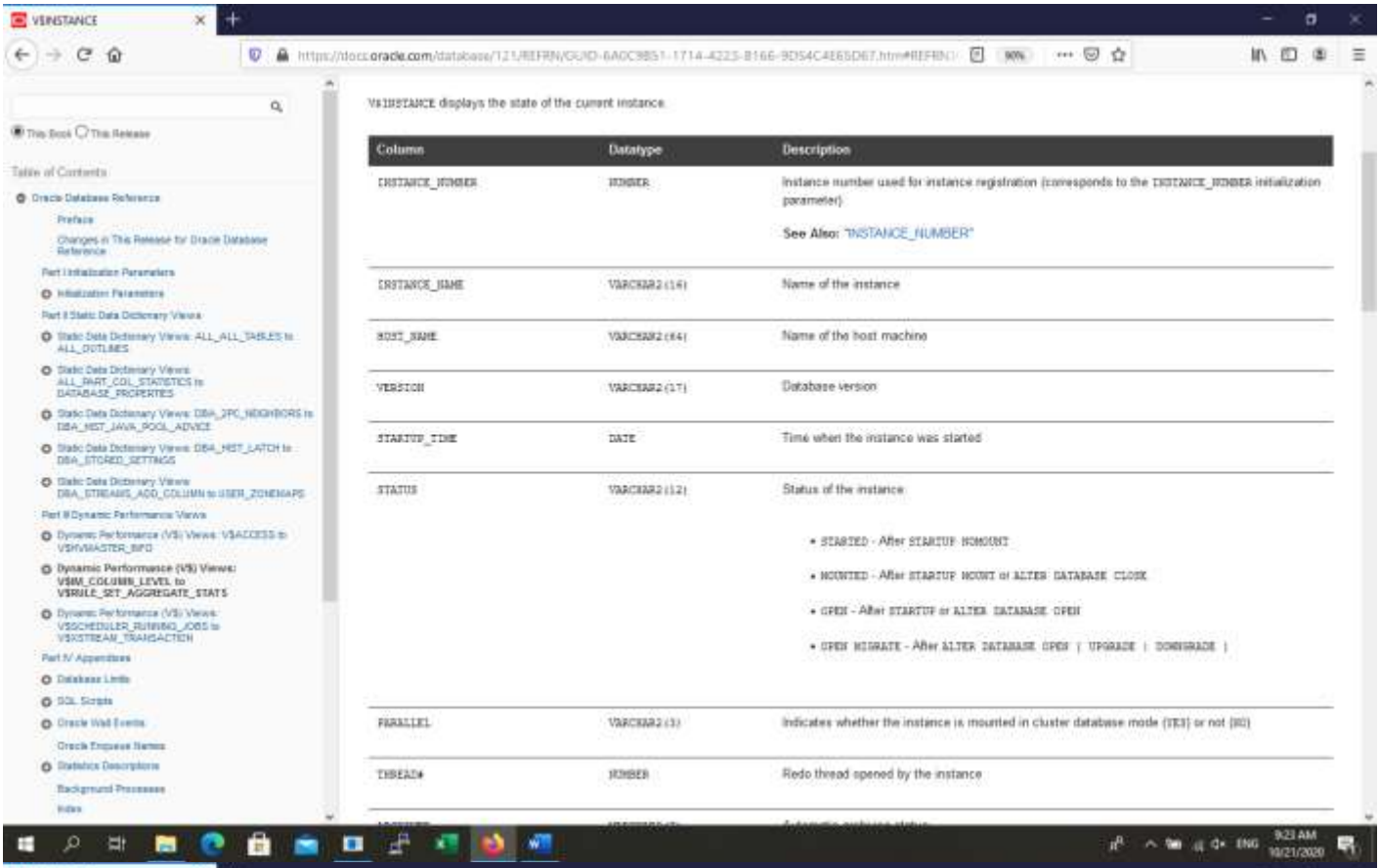
```
SQL> conn test
Enter password:
ERROR:
ORA-01035: ORACLE only available to users with RESTRICTED SESSION privilege
```

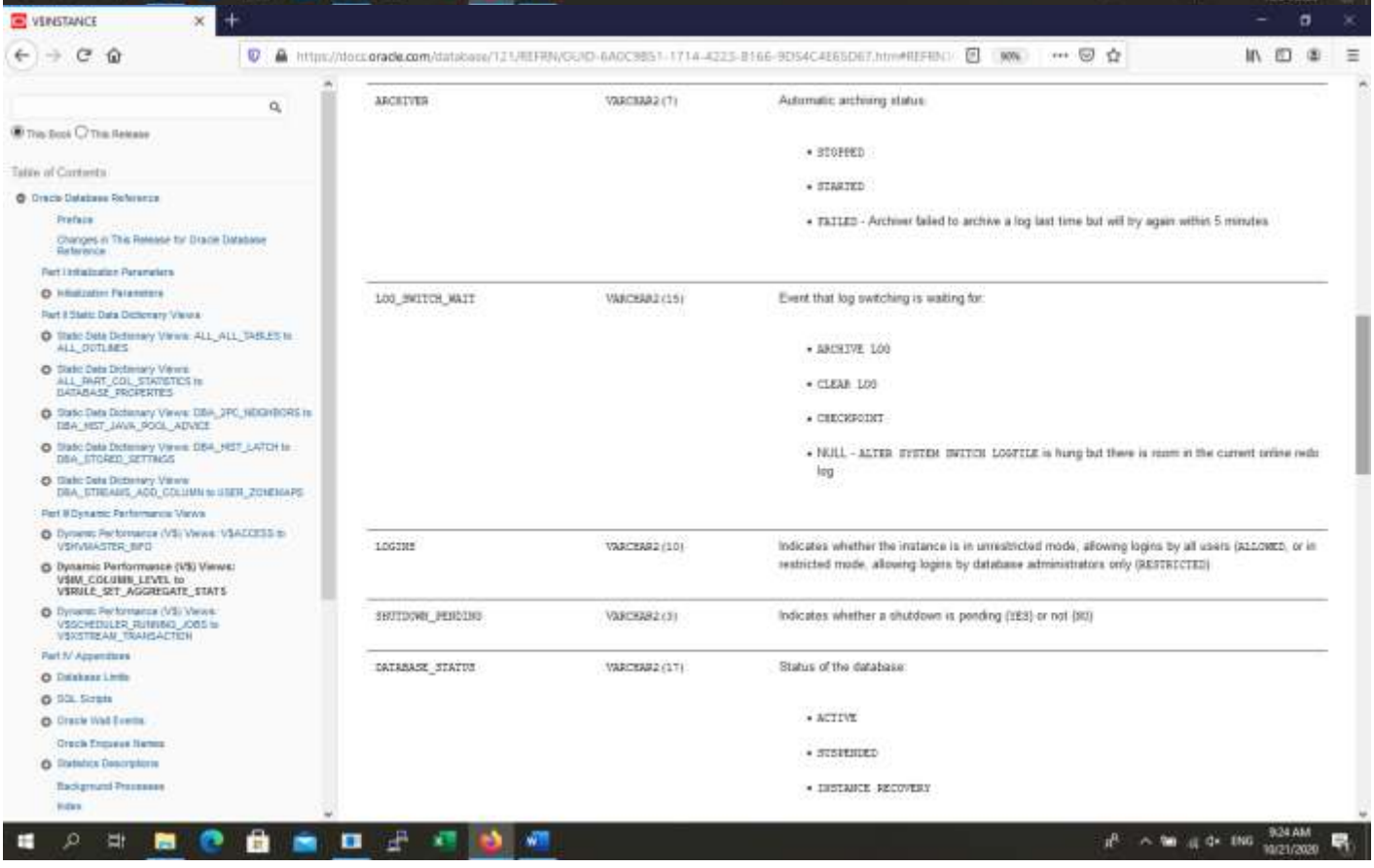
Obs. Daca instalarea/crearea bazei de date s-a facut cu optiunea *Create Container* atunci in userul SYS trebuie acordat urmatorul privilegiu userului C##ABD1 (chiar daca are privilegiul de DBA) :

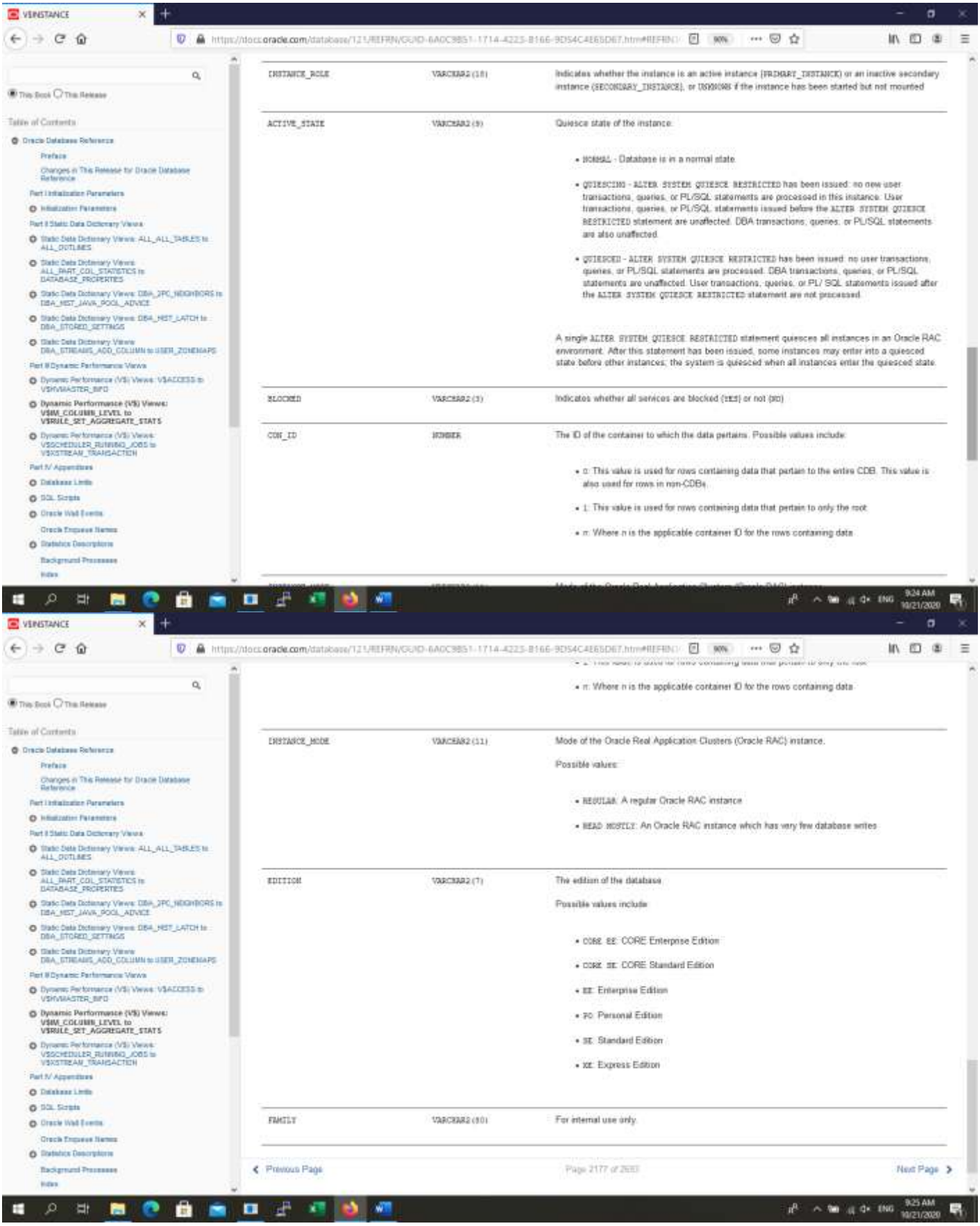
```
SQL> grant create user to C##ABD1 container=all ;
```

```
SQL> desc V$INSTANCE;
```

Name	Null?	Type
INSTANCE_NUMBER		NUMBER
INSTANCE_NAME		VARCHAR2(16)
HOST_NAME		VARCHAR2(64)
VERSION		VARCHAR2(17)
STARTUP_TIME		DATE
STATUS		VARCHAR2(12)
PARALLEL		VARCHAR2(3)
THREAD#		NUMBER
ARCHIVER		VARCHAR2(7)
LOG_SWITCH_WAIT		VARCHAR2(15)
LOGINS		VARCHAR2(10)
SHUTDOWN_PENDING		VARCHAR2(3)
DATABASE_STATUS		VARCHAR2(17)
INSTANCE_ROLE		VARCHAR2(18)
ACTIVE_STATE		VARCHAR2(9)
BLOCKED		VARCHAR2(3)
CON_ID		NUMBER
INSTANCE_MODE		VARCHAR2(11)
EDITION		VARCHAR2(7)
FAMILY		VARCHAR2(80)


V$INSTANCE
https://docs.oracle.com/database/121/REFRN/GUID-6A0C9B51-1714-4223-B166-9D54C4E65D67.htm#REFRN
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V$INSTANCE displays the state of the current instance.
Column
Datatype
Description
INSTANCE_NUMBER
NUMBER
Instance number used for instance registration (corresponds to the INSTANCE_NUMBER initialization parameter)
See Also: "INSTANCE_NUMBER"
INSTANCE_NAME
VARCHAR2 (16)
Name of the instance
HOST_NAME
VARCHAR2 (64)
Name of the host machine
VERSION
VARCHAR2 (17)
Database version
STARTUP_TIME
DATE
Time when the instance was started
STATUS
VARCHAR2 (12)
Status of the instance
STARTED - After STARTUP NOMOUNT
MOUNTED - After STARTUP MOUNT or ALTER DATABASE CLOSE
OPEN - After STARTUP or ALTER DATABASE OPEN
OPEN MIGRATE - After ALTER DATABASE OPEN { UPGRADE | DOWNGRADE }
PARALLEL
VARCHAR2 (3)
Indicates whether the instance is mounted in cluster database mode (YES) or not (NO)
THREAD#
NUMBER
Redo thread opened by the instance
9:23 AM
10/21/2020


V$INSTANCE
https://docs.oracle.com/database/121/REFRN/GUID-6A0C9B51-1714-4223-B166-9D54C4E65D67.htm#REFRN
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ARCHIVER
VARCHAR2 (7)
Automatic archiving status:
STOPPED
STARTED
FAILED - Archiver failed to archive a log last time but will try again within 5 minutes
LOG_SWITCH_WAIT
VARCHAR2 (15)
Event that log switching is waiting for:
ARCHIVE LOG
CLEAR LOG
CHECKPOINT
NULL - ALTER SYSTEM SWITCH LOGFILE is hung but there is room in the current online redo log
LOGINS
VARCHAR2 (10)
Indicates whether the instance is in unrestricted mode, allowing logins by all users (ALLOWED), or in restricted mode, allowing logins by database administrators only (RESTRICTED)
SHUTDOWN_PENDING
VARCHAR2 (3)
Indicates whether a shutdown is pending (YES) or not (NO)
DATABASE_STATUS
VARCHAR2 (17)
Status of the database:
ACTIVE
SUSPENDED
INSTANCE RECOVERY
9:24 AM
10/21/2020


VSINSTANCE
https://docs.oracle.com/database/121/REFRN/GUID-6A0C9B51-1714-4223-B166-9D54C4E65D67.htm#REFRN
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INSTANCE_ROLE
VARCHAR2 (18)
Indicates whether the instance is an active instance (PRIMARY_INSTANCE) or an inactive secondary instance (SECONDARY_INSTANCE), or UNKNOWN if the instance has been started but not mounted
ACTIVE_STATE
VARCHAR2 (9)
Quiesce state of the instance:
NORMAL - Database is in a normal state
QUIESCING - ALTER SYSTEM QUIESCE RESTRICTED has been issued: no new user transactions, queries, or PL/SQL statements are processed in this instance. User transactions, queries, or PL/SQL statements issued before the ALTER SYSTEM QUIESCE RESTRICTED statement are unaffected. DBA transactions, queries, or PL/SQL statements are also unaffected.
QUIESCED - ALTER SYSTEM QUIESCE RESTRICTED has been issued: no user transactions, queries, or PL/SQL statements are processed. DBA transactions, queries, or PL/SQL statements are unaffected. User transactions, queries, or PL/ SQL statements issued after the ALTER SYSTEM QUIESCE RESTRICTED statement are not processed.
A single ALTER SYSTEM QUIESCE RESTRICTED statement quiesces all instances in an Oracle RAC environment. After this statement has been issued, some instances may enter into a quiesced state before other instances; the system is quiesced when all instances enter the quiesced state.
BLOCKED
VARCHAR2 (3)
Indicates whether all services are blocked (YES) or not (NO)
CON_ID
NUMBER
The ID of the container to which the data pertains. Possible values include:
0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs.
1: This value is used for rows containing data that pertain to only the root
n: Where n is the applicable container ID for the rows containing data
INSTANCE_MODE
VARCHAR2 (11)
Mode of the Oracle Real Application Clusters (Oracle RAC) instance.
Possible values:
REGULAR: A regular Oracle RAC instance
READ MOSTLY: An Oracle RAC instance which has very few database writes
EDITION
VARCHAR2 (7)
The edition of the database
Possible values include:
CORE EE: CORE Enterprise Edition
CORE SE: CORE Standard Edition
EE: Enterprise Edition
PE: Personal Edition
SE: Standard Edition
XE: Express Edition
FAMILY
VARCHAR2 (30)
For internal use only
Previous Page
Next Page

```
SQL> select logins from v$instance;

LOGINS
----------
RESTRICTED

SQL> alter system disable restricted session;

System altered.

SQL> select logins from v$instance;

LOGINS
----------
ALLOWED

SQL> conn test
Enter password: ***
Connected.
```

Exercitii (Cap.15 si Cap.16) :

1. Sa se creeze userul STUD1 si sa i se aloce un spatiu de 2 Mb pentru a crea propriile obiecte.

2. Verificati in dictionar care sunt tablespace-urile permanente si temporare asignate userului STUD1.

3. Sa se arate din dictionar care sunt privilegiile de sistem acordate userului STUD1.

4. Sa i se acorde userului STUD1 privilegii de a insera si vizualiza date in tabela EMP din schema ABD1.

5. Sa arate din dictionar care sunt userii grantificati de catre userul ABD1, care sunt privilegiile acordate si pe ce tabele sunt acordate.